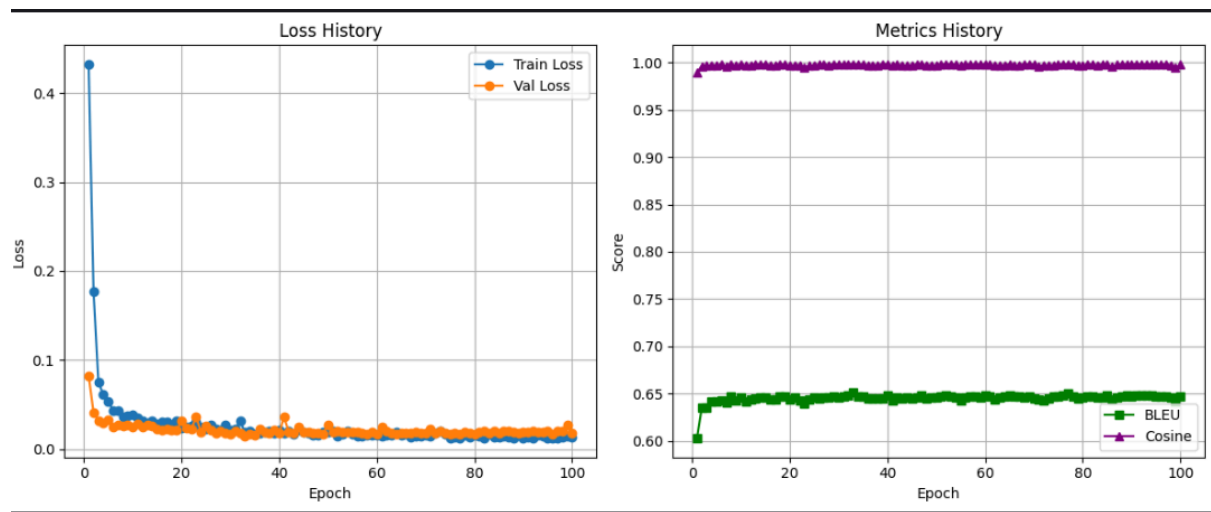


3_WORDS_RESULTS



epoch	train_loss	val_loss	bleu	cosine
1	0.432654	0.082592	0.602335	0.989161
2	0.176599	0.041134	0.635153	0.996604
3	0.075105	0.031796	0.634502	0.996653
4	0.061233	0.029638	0.641114	0.997563
5	0.05395	0.033246	0.641053	0.997265
6	0.043321	0.025312	0.642303	0.998061
7	0.043519	0.027643	0.640329	0.996496
8	0.035788	0.025952	0.647404	0.998161
9	0.037917	0.027335	0.642316	0.996893
10	0.038294	0.024512	0.646335	0.998228
11	0.034701	0.028777	0.641505	0.997165
12	0.032217	0.024665	0.643724	0.997534
13	0.029294	0.027004	0.644914	0.997837
14	0.032173	0.025706	0.6453	0.998047
15	0.028844	0.022821	0.645978	0.997946
16	0.03011	0.021755	0.643933	0.99734
17	0.030922	0.022351	0.643688	0.997341

18	0.02709	0.021316	0.646524	0.998135
19	0.031689	0.020935	0.64717	0.997928
20	0.024105	0.031724	0.644054	0.996893
21	0.024525	0.023452	0.646267	0.997566
22	0.025981	0.022155	0.643067	0.997538
23	0.03046	0.035839	0.638922	0.99512
24	0.024686	0.019255	0.643555	0.996928
25	0.022893	0.026396	0.645897	0.997406
26	0.027179	0.021159	0.64462	0.997752
27	0.022272	0.017919	0.645551	0.997975
28	0.021408	0.019967	0.64563	0.997593
29	0.026615	0.018281	0.646577	0.998275
30	0.020911	0.017406	0.646035	0.998137
31	0.022619	0.020151	0.646922	0.997831
32	0.03129	0.017448	0.648491	0.998506
33	0.018896	0.014983	0.650815	0.998321
34	0.020247	0.016468	0.646791	0.997836
35	0.017262	0.015802	0.647284	0.997959
36	0.018452	0.022841	0.644892	0.997349
37	0.018682	0.019607	0.644919	0.997544
38	0.020464	0.0185	0.644903	0.997504
39	0.018446	0.020926	0.644755	0.998071
40	0.020929	0.017454	0.647504	0.998353
41	0.018329	0.03631	0.642467	0.996904
42	0.020639	0.018862	0.64577	0.99787
43	0.016609	0.017922	0.64484	0.997298
44	0.021558	0.024515	0.645348	0.99724

45	0.018657	0.018995	0.644367	0.997375
46	0.017449	0.018857	0.64573	0.998125
47	0.015203	0.017886	0.648124	0.99836
48	0.015865	0.018039	0.644488	0.99723
49	0.019316	0.017372	0.645852	0.997649
50	0.018815	0.0273	0.645939	0.997214
51	0.020285	0.019301	0.646415	0.998091
52	0.014145	0.02059	0.647481	0.998322
53	0.017132	0.019288	0.647325	0.998014
54	0.020196	0.019585	0.645313	0.998016
55	0.017247	0.019672	0.642406	0.996835
56	0.014876	0.018713	0.646135	0.99798
57	0.014591	0.017861	0.647401	0.997903
58	0.01621	0.017215	0.646801	0.997823
59	0.016047	0.018617	0.646149	0.997819
60	0.015211	0.016924	0.648088	0.998337
61	0.014575	0.024753	0.646902	0.997814
62	0.015375	0.02057	0.643557	0.997147
63	0.015636	0.018141	0.646119	0.997452
64	0.019479	0.017042	0.64668	0.997452
65	0.015895	0.017798	0.647862	0.998328
66	0.01719	0.017674	0.646913	0.996961
67	0.013151	0.018159	0.646911	0.997653
68	0.01511	0.018632	0.646116	0.998105
69	0.014543	0.01843	0.646414	0.998056
70	0.016271	0.018541	0.644772	0.997975
71	0.014496	0.022627	0.643432	0.996198

72	0.017537	0.018049	0.642308	0.997454
73	0.020105	0.019823	0.644288	0.997521
74	0.016647	0.017454	0.647072	0.997472
75	0.012318	0.017188	0.647171	0.997892
76	0.013096	0.017834	0.647662	0.998096
77	0.012831	0.017075	0.649744	0.998532
78	0.015634	0.019363	0.647082	0.998124
79	0.013232	0.017619	0.644343	0.996686
80	0.015988	0.017252	0.646318	0.997388
81	0.014274	0.018725	0.64691	0.997949
82	0.012033	0.020304	0.647077	0.997942
83	0.01691	0.018357	0.645622	0.997403
84	0.013949	0.019941	0.645747	0.998206
85	0.013478	0.01845	0.647718	0.99806
86	0.014613	0.019771	0.64413	0.995623
87	0.013044	0.020563	0.645544	0.997881
88	0.012223	0.019474	0.646746	0.997948
89	0.016384	0.017704	0.647429	0.997721
90	0.012767	0.018936	0.646625	0.998125
91	0.013011	0.019376	0.647446	0.998312
92	0.011906	0.020089	0.647456	0.998138
93	0.015299	0.019451	0.648193	0.998117
94	0.014901	0.019606	0.647972	0.998045
95	0.012845	0.020391	0.646602	0.997837
96	0.01281	0.017418	0.646671	0.997835
97	0.012096	0.019985	0.646863	0.99825
98	0.012972	0.020176	0.64633	0.997529

99	0.0147	0.02704	0.641224	0.994707
100	0.013751	0.017683	0.646611	0.997785

```
=== SAMPLE RECONSTRUCTIONS (BEST MODEL) ===
```

```
Input: hill behind mountain
```

```
Target: hill behind mountain
```

```
Pred: hill behind mountain
```

```
-----
```

```
Input: king makes offering
```

```
Target: king makes offering
```

```
Pred: king makes offering
```

```
-----
```

```
Input: speaks king beauty good /
```

```
Target: king speaks beauty / good
```

```
Pred: king speaks beauty / good
```

```
-----
```

```
Input: town sees king
```

```
Target: king sees town
```

```
Pred: king sees town
```

```
-----
```

```
Input: reed servant sees shelter
```

```
Target: servant sees reed shelter
```

```
Pred: servant sees reed shelter
```

```
-----
```

```
Input: work does hand
```

```
Target: hand does work
```

```
Pred: hand does work
```

```
-----
```

```
Input: meets lake river
```

```
Target: river meets lake
```

```
Pred: river meets lake
```

```
-----
```

```
Input: people holds city
```

```
Target: city holds people
```

```
Pred: city holds people
```

```
-----
```

```
Input: woman seated listens
```

```
Target: seated woman listens
```

```
Pred: seated woman listens
```

```
-----
```

```
Input: bread servant of gives loaf
```

```
Target: servant gives loaf of bread
```

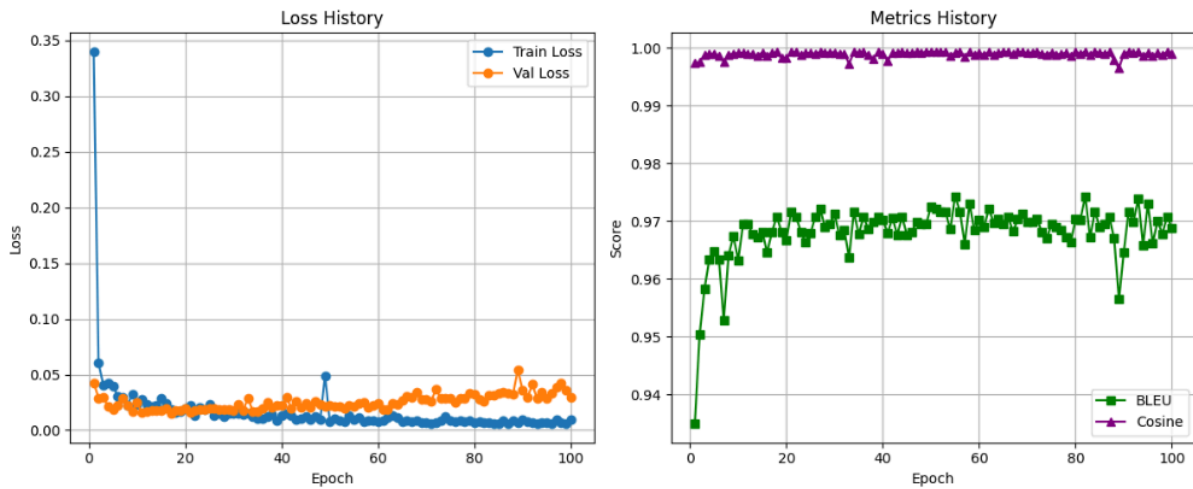
```
Pred: servant gives loaf of bread
```

```
-----
```

```
Training complete. Best model saved in: ./best_model_checkpoint
```

<https://www.kaggle.com/code/ahmed3khaled/bart-3-words-english>

4_WORDS_RESULTS



epoch	train_loss	val_loss	bleu	cosine
1	0.339855	0.042683	0.934855	0.997292
2	0.060421	0.029004	0.950305	0.997498
3	0.040521	0.029398	0.958195	0.998787
4	0.042064	0.021449	0.963309	0.998955
5	0.039886	0.019023	0.964664	0.998981
6	0.030157	0.021907	0.963427	0.998672
7	0.029501	0.028836	0.952796	0.997495
8	0.022415	0.022599	0.964092	0.998753
9	0.032027	0.017209	0.967335	0.999031
10	0.023041	0.024743	0.963234	0.99906
11	0.027971	0.015611	0.969444	0.999161
12	0.023369	0.017084	0.969503	0.999003
13	0.021179	0.017481	0.967732	0.998904
14	0.021886	0.017683	0.967217	0.998642
15	0.028493	0.017366	0.968129	0.999188
16	0.023971	0.019457	0.964526	0.998633
17	0.018592	0.014569	0.968081	0.999136

18	0.01551	0.017595	0.970773	0.999223
19	0.016577	0.018116	0.968075	0.998174
20	0.019321	0.019456	0.966697	0.998216
21	0.02212	0.016244	0.971654	0.99924
22	0.013149	0.017662	0.970724	0.999234
23	0.02062	0.018325	0.968147	0.998722
24	0.018818	0.018268	0.96631	0.999083
25	0.023052	0.019983	0.967846	0.999082
26	0.013038	0.019958	0.970758	0.998964
27	0.015281	0.018525	0.972095	0.999285
28	0.012515	0.018639	0.968962	0.999186
29	0.014613	0.018473	0.96951	0.999195
30	0.015057	0.017445	0.971165	0.999221
31	0.014974	0.023579	0.967516	0.998911
32	0.014419	0.017523	0.968513	0.999
33	0.01536	0.028671	0.963697	0.997215
34	0.01264	0.017227	0.971606	0.999265
35	0.010405	0.016818	0.96764	0.999175
36	0.010232	0.019847	0.970724	0.999234
37	0.012	0.025317	0.968606	0.998762
38	0.013862	0.020389	0.969842	0.998076
39	0.008721	0.022431	0.970773	0.999259
40	0.013443	0.021874	0.970268	0.999014
41	0.015281	0.029513	0.967875	0.997729
42	0.013519	0.019962	0.9705	0.999055
43	0.009136	0.025987	0.967632	0.999137
44	0.010593	0.020148	0.970724	0.999258

45	0.012133	0.024305	0.967632	0.999089
46	0.009329	0.02025	0.968073	0.999086
47	0.01205	0.02594	0.969892	0.999282
48	0.009541	0.023307	0.969444	0.999189
49	0.048955	0.021722	0.969451	0.999234
50	0.007441	0.022668	0.972536	0.999316
51	0.010492	0.020998	0.972087	0.999262
52	0.008571	0.021013	0.971606	0.99926
53	0.007452	0.019915	0.971606	0.999277
54	0.012915	0.022664	0.968539	0.998565
55	0.009182	0.020936	0.974283	0.999172
56	0.011114	0.023637	0.971647	0.999263
57	0.008133	0.025313	0.966003	0.998513
58	0.008901	0.020684	0.973025	0.99931
59	0.008923	0.021923	0.968397	0.998761
60	0.007537	0.023716	0.970268	0.999047
61	0.008314	0.018456	0.968939	0.998838
62	0.011308	0.018479	0.972087	0.99923
63	0.012847	0.023714	0.969711	0.998844
64	0.011424	0.022932	0.970276	0.99919
65	0.007509	0.026787	0.969402	0.999238
66	0.008378	0.030856	0.970724	0.999268
67	0.008073	0.029439	0.968165	0.998874
68	0.008525	0.033694	0.970284	0.999253
69	0.007061	0.027625	0.971214	0.99925
70	0.00704	0.027557	0.969828	0.999123
71	0.006298	0.02569	0.969884	0.999196

72	0.006548	0.036962	0.970284	0.999251
73	0.008444	0.028779	0.968098	0.998992
74	0.011908	0.028876	0.966959	0.998863
75	0.008365	0.028971	0.969395	0.999164
76	0.00783	0.025265	0.96898	0.998842
77	0.00903	0.02898	0.968501	0.99899
78	0.007427	0.028911	0.967192	0.999098
79	0.008254	0.032787	0.966395	0.998681
80	0.006933	0.032372	0.970284	0.999236
81	0.007923	0.027384	0.970208	0.998957
82	0.006935	0.025553	0.974298	0.99936
83	0.006678	0.031039	0.967168	0.998759
84	0.005925	0.031237	0.971647	0.999253
85	0.005592	0.033156	0.968905	0.99919
86	0.008951	0.033988	0.969436	0.999023
87	0.006318	0.033066	0.970717	0.999224
88	0.0085	0.032186	0.966945	0.997821
89	0.007214	0.054395	0.956434	0.996427
90	0.009116	0.035672	0.964597	0.999032
91	0.007416	0.029342	0.971654	0.999287
92	0.006693	0.041064	0.969828	0.999122
93	0.00602	0.028243	0.97385	0.999263
94	0.006656	0.033761	0.965831	0.998574
95	0.006387	0.029071	0.972961	0.999148
96	0.00562	0.033228	0.966163	0.998666
97	0.009182	0.038366	0.969992	0.999069
98	0.006429	0.042225	0.967774	0.998803

99	0.005962	0.03599	0.970773	0.999276
100	0.009362	0.029642	0.968741	0.998897

=== SAMPLE RECONSTRUCTIONS (BEST MODEL) ===

Input: gate god temple watch

Target: god watch temple gate

Pred: god watch temple gate

Input: open town gate servant

Target: servant open town gate

Pred: servant open town gate

Input: youth heart seek sweet

Target: youth seek sweet heart

Pred: youth seek sweet heart

Input: path people walk desert

Target: people walk desert path

Pred: people walk desert path

Input: people build town house

Target: people build house town

Pred: people build house town

Input: road youth father teach

Target: father teach youth road

Pred: father teach youth road

Input: man lake drink water

Target: man drink lake water

Pred: man drink lake water

Input: father road youth teach

Target: father teach youth road

Pred: father teach youth road

Input: people god river protect

Target: god protect river people

Pred: god protect river people

Input: finds stone man brick

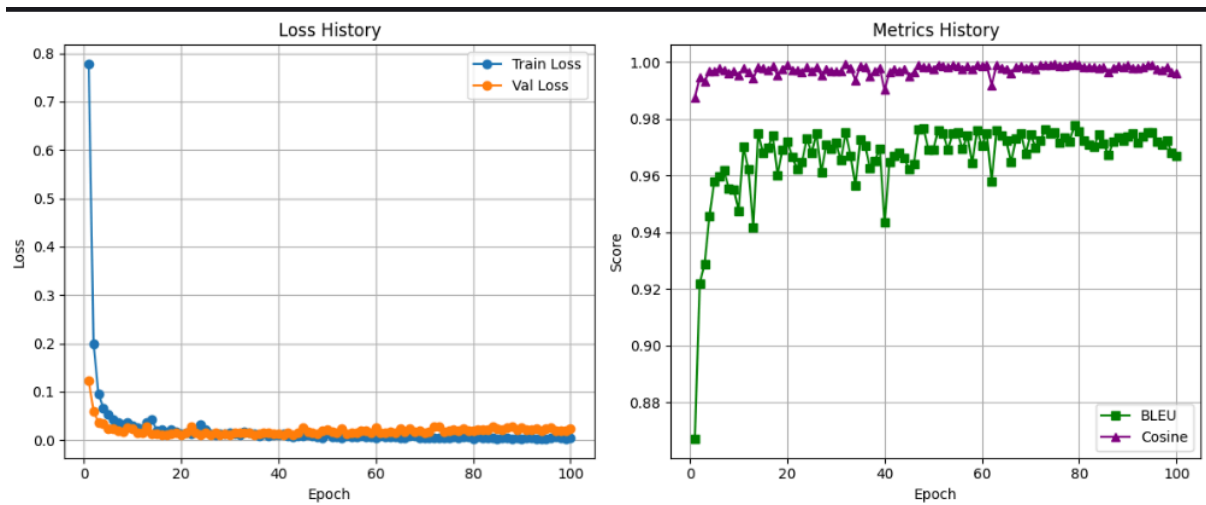
Target: man finds stone brick

Pred: man finds stone brick

Training complete. Best model saved in: `./best_model_checkpoint`

<https://www.kaggle.com/code/immortusdeimmortal/bart-4-words-english>

5_WORDS_RESULTS



epoch	train_loss	val_loss	bleu	cosine
1	0.778475	0.12329	0.866998	0.987499
2	0.199693	0.060895	0.921944	0.994448
3	0.096169	0.036547	0.928805	0.99324
4	0.067022	0.035248	0.945766	0.996591
5	0.053562	0.024301	0.958026	0.996777
6	0.042082	0.024157	0.959827	0.997708
7	0.036942	0.019212	0.961665	0.997264
8	0.032066	0.018664	0.955326	0.996023
9	0.036086	0.025603	0.955078	0.996931
10	0.030831	0.023364	0.947586	0.99537
11	0.026756	0.014924	0.970044	0.997729
12	0.024596	0.015954	0.962139	0.996566
13	0.036874	0.028031	0.9418	0.994126
14	0.042003	0.012707	0.974843	0.998273
15	0.019768	0.012362	0.96801	0.997867
16	0.02142	0.011177	0.969664	0.997006
17	0.015587	0.0111	0.97406	0.99842

18	0.0224	0.012573	0.960124	0.995429
19	0.017977	0.015223	0.969131	0.997602
20	0.013086	0.011753	0.971829	0.998796
21	0.014761	0.01534	0.966681	0.996968
22	0.012803	0.027289	0.962182	0.99724
23	0.021792	0.015003	0.964636	0.996466
24	0.032392	0.012071	0.972866	0.998162
25	0.021462	0.015819	0.967969	0.996701
26	0.012457	0.011293	0.974793	0.998269
27	0.011011	0.015744	0.960978	0.995236
28	0.012514	0.01124	0.970965	0.997635
29	0.014088	0.012369	0.969222	0.996922
30	0.01626	0.011699	0.971611	0.996617
31	0.012982	0.014471	0.965258	0.996621
32	0.01175	0.016221	0.974984	0.99915
33	0.018272	0.01553	0.966894	0.997894
34	0.015466	0.015473	0.956346	0.993497
35	0.012706	0.011882	0.972461	0.998729
36	0.00859	0.014099	0.970443	0.998235
37	0.013327	0.014752	0.962373	0.994869
38	0.009919	0.015769	0.965119	0.996655
39	0.011987	0.013712	0.969436	0.997888
40	0.010543	0.013772	0.943323	0.990108
41	0.012544	0.011954	0.964804	0.996559
42	0.009259	0.015877	0.967047	0.997527
43	0.007238	0.011249	0.967889	0.996713
44	0.008548	0.015062	0.966106	0.997573

45	0.009124	0.025347	0.962245	0.995133
46	0.010833	0.017484	0.963898	0.996363
47	0.008547	0.015898	0.976339	0.998855
48	0.007636	0.013583	0.976535	0.998172
49	0.004737	0.019595	0.96904	0.998127
50	0.014203	0.021819	0.969038	0.997377
51	0.007975	0.017338	0.975864	0.999048
52	0.006434	0.014643	0.974857	0.998434
53	0.005545	0.023144	0.969138	0.998085
54	0.009947	0.014015	0.974756	0.998813
55	0.007642	0.015733	0.975047	0.99861
56	0.007581	0.015598	0.96922	0.997565
57	0.008542	0.020189	0.974122	0.99858
58	0.007444	0.019685	0.964201	0.997489
59	0.006483	0.016345	0.975983	0.998894
60	0.004005	0.026369	0.970356	0.998414
61	0.009468	0.016209	0.974851	0.998751
62	0.007676	0.01545	0.957762	0.991772
63	0.007621	0.016643	0.976024	0.998859
64	0.007581	0.01581	0.974202	0.997983
65	0.004328	0.024589	0.972246	0.997619
66	0.005298	0.01803	0.964579	0.996001
67	0.008307	0.022978	0.972901	0.99831
68	0.008984	0.017435	0.974736	0.998494
69	0.005819	0.019425	0.967699	0.997962
70	0.004889	0.015425	0.974367	0.998487
71	0.005445	0.018004	0.969927	0.997342

72	0.004482	0.027656	0.972139	0.99894
73	0.005815	0.027249	0.976198	0.998895
74	0.004434	0.018295	0.974697	0.99877
75	0.00549	0.019525	0.975235	0.999125
76	0.007703	0.021453	0.971409	0.99844
77	0.004381	0.018735	0.973421	0.998714
78	0.007242	0.024178	0.971913	0.999004
79	0.006865	0.020534	0.977623	0.999387
80	0.002861	0.01893	0.975434	0.998897
81	0.006129	0.021405	0.972284	0.998239
82	0.00418	0.021062	0.970866	0.998311
83	0.003895	0.021887	0.969948	0.998357
84	0.003955	0.029099	0.974573	0.997871
85	0.00365	0.023934	0.971335	0.998366
86	0.005868	0.021349	0.967087	0.996384
87	0.005489	0.025452	0.971812	0.997929
88	0.003422	0.027585	0.97326	0.998634
89	0.004776	0.021776	0.972196	0.998138
90	0.00275	0.026547	0.973771	0.998853
91	0.004619	0.022292	0.974653	0.998023
92	0.005541	0.022306	0.971621	0.997918
93	0.002118	0.024991	0.973859	0.998122
94	0.003264	0.020496	0.975229	0.998819
95	0.002249	0.024974	0.975072	0.999037
96	0.005195	0.025464	0.971911	0.997484
97	0.006134	0.020002	0.970876	0.997117
98	0.005198	0.020783	0.972394	0.998337

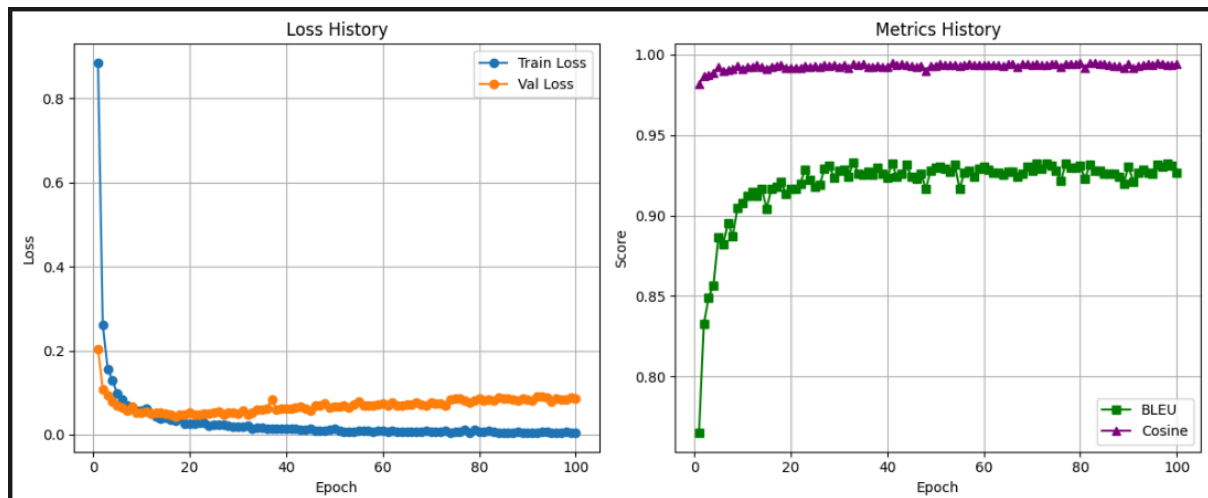
99	0.003049	0.020075	0.967885	0.996313
100	0.005206	0.02395	0.966753	0.995893

```
=== SAMPLE RECONSTRUCTIONS (BEST MODEL) ===
Input:  ripple water drinks woman
Target: woman drinks water ripple
Pred:   woman drinks water ripple
-----
Input:  emerge ripple water snake lake underground
Target: underground lake water ripple snake emerge
Pred:   underground lake water ripple snake emerge
-----
Input:  woman bread altar offers
Target: woman offers bread altar
Pred:   woman offers bread altar
-----
Input:  strength power force god grants
Target: god grants power strength force
Pred:   god grants power strength force
-----
Input:  plows man great strong field
Target: strong man plows great field
Pred:   strong man plows great field
-----
Input:  become / sweet good beautiful plant (ankh) lotus life
Target: beautiful / good lotus plant become sweet life (ankh)
Pred:   beautiful / good lotus plant become sweet life (ankh)
-----
Input:  (or road opens / exists is found) town
Target: opens / exists (or is found) road town
Pred:   opens / exists (or is found) road town
-----
Input:  man beautiful old becomes youth
Target: beautiful youth becomes old man
Pred:   beautiful youth becomes old man
-----
Input:  field (tills) great / mighty plows sickle
Target: sickle plows (tills) great / mighty field
Pred:   sickle plows (tills) great / mighty field
-----
Input:  king servant commands
Target: king commands servant
Pred:   king commands servant
-----

Training complete. Best model saved in: ./best_model_checkpoint
```

<https://www.kaggle.com/code/demaciandedemacia/n/bart-5-words-english>

6_WORDS_RESULTS



epoch	train_loss	val_loss	bleu	cosine
1	0.88511	0.204822	0.764889	0.981602
2	0.260333	0.108655	0.832848	0.986814
3	0.156556	0.093891	0.849262	0.987333
4	0.129921	0.0802	0.85676	0.988886
5	0.097358	0.069079	0.886644	0.992096
6	0.083589	0.064302	0.882078	0.989603
7	0.070591	0.056586	0.895376	0.990518
8	0.065528	0.066616	0.887144	0.991184
9	0.057798	0.053341	0.90501	0.993146
10	0.056937	0.053086	0.907539	0.990916
11	0.061406	0.054267	0.912379	0.992504
12	0.050853	0.05043	0.914794	0.992656
13	0.042	0.051673	0.9123	0.993308
14	0.037837	0.052205	0.91673	0.992354
15	0.041248	0.050743	0.90418	0.991415
16	0.03591	0.048398	0.916531	0.992552
17	0.032524	0.044256	0.917965	0.993293

18	0.039941	0.047184	0.92124	0.993495
19	0.026487	0.04683	0.913293	0.991924
20	0.027311	0.05184	0.916821	0.99195
21	0.026531	0.048441	0.916431	0.991718
22	0.029774	0.048731	0.919954	0.991892
23	0.028264	0.050932	0.928796	0.993227
24	0.022391	0.050719	0.921989	0.992365
25	0.024989	0.052068	0.917571	0.992715
26	0.023	0.055752	0.919027	0.992213
27	0.022785	0.048627	0.929233	0.993688
28	0.020792	0.051702	0.930715	0.992957
29	0.020213	0.052193	0.923336	0.993691
30	0.018622	0.049408	0.927951	0.992658
31	0.020108	0.056734	0.9287	0.993812
32	0.022606	0.048523	0.924296	0.991525
33	0.013599	0.051563	0.933133	0.993979
34	0.016053	0.058756	0.926004	0.993316
35	0.017406	0.059037	0.925515	0.993967
36	0.015343	0.061757	0.927501	0.9924
37	0.015231	0.084767	0.925095	0.992565
38	0.01482	0.059577	0.929895	0.993134
39	0.013299	0.062168	0.925962	0.992603
40	0.013558	0.063128	0.923387	0.992241
41	0.014089	0.06138	0.932531	0.99463
42	0.014332	0.063638	0.924058	0.99372
43	0.0123	0.067698	0.926051	0.993955
44	0.011923	0.062185	0.931579	0.993561

45	0.013553	0.058264	0.924235	0.992724
46	0.009577	0.069796	0.922766	0.992623
47	0.009787	0.070478	0.926205	0.993182
48	0.010225	0.075149	0.916313	0.990049
49	0.012524	0.065888	0.928096	0.993167
50	0.013811	0.068195	0.929591	0.993119
51	0.010136	0.0668	0.930099	0.99403
52	0.007876	0.068994	0.928905	0.993547
53	0.007706	0.063721	0.927482	0.993713
54	0.00732	0.071431	0.931543	0.993592
55	0.009589	0.079428	0.916409	0.992752
56	0.009514	0.070638	0.926902	0.99369
57	0.009464	0.069707	0.927733	0.994365
58	0.008003	0.069485	0.923818	0.993378
59	0.0105	0.072867	0.928826	0.99378
60	0.00991	0.07324	0.930422	0.99355
61	0.006483	0.069167	0.928536	0.993561
62	0.009324	0.075945	0.926854	0.993791
63	0.008043	0.069541	0.926845	0.993453
64	0.006795	0.068858	0.925524	0.992989
65	0.007946	0.072835	0.927347	0.993948
66	0.006672	0.072509	0.927372	0.994201
67	0.006262	0.077545	0.923933	0.992059
68	0.006646	0.071417	0.925857	0.994296
69	0.009328	0.06951	0.930605	0.994135
70	0.006758	0.076617	0.928147	0.993368
71	0.006209	0.074405	0.932161	0.993964

72	0.00733	0.07455	0.928833	0.993303
73	0.008668	0.068505	0.932107	0.993642
74	0.005783	0.082933	0.930907	0.994194
75	0.006409	0.086174	0.927629	0.994083
76	0.0072	0.085161	0.921892	0.992616
77	0.011103	0.081885	0.932331	0.99446
78	0.005545	0.077454	0.929864	0.994483
79	0.01097	0.082289	0.929454	0.993925
80	0.00642	0.085824	0.930751	0.994819
81	0.006491	0.081254	0.922854	0.991952
82	0.009696	0.083909	0.931931	0.995039
83	0.006884	0.080978	0.927866	0.994609
84	0.004435	0.087795	0.927921	0.994479
85	0.005148	0.085384	0.926284	0.994146
86	0.004374	0.086149	0.925985	0.993483
87	0.004629	0.083413	0.925748	0.993237
88	0.007689	0.080708	0.924314	0.993015
89	0.004716	0.086482	0.9199	0.991943
90	0.005377	0.084483	0.930512	0.993991
91	0.005	0.082046	0.920956	0.99176
92	0.004538	0.091793	0.926563	0.992746
93	0.00728	0.091344	0.928271	0.993555
94	0.007591	0.088246	0.926832	0.994114
95	0.003971	0.080113	0.925852	0.993559
96	0.005776	0.08673	0.931398	0.995001
97	0.004613	0.083648	0.930209	0.994434
98	0.006556	0.084901	0.932254	0.993874

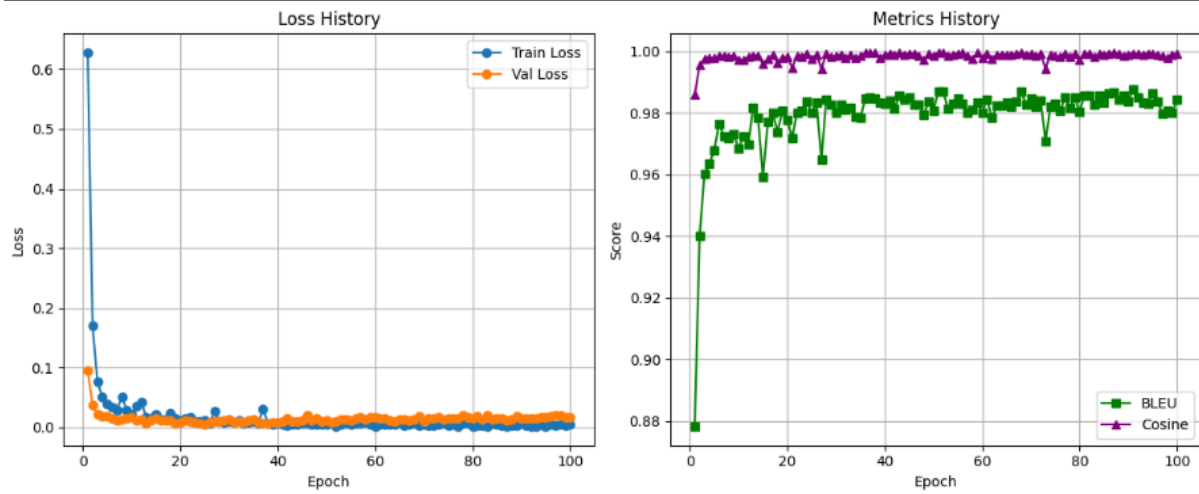
99	0.004714	0.088609	0.930946	0.993491
100	0.005523	0.085852	0.92668	0.99443

```
=== SAMPLE RECONSTRUCTIONS (BEST MODEL) ===
Input:  good beautiful negation / / make (no/not) do fear heart thing
Target: negation (no/not) do / make thing beautiful / good heart fear
Pred:   negation (no/not) do / make thing beautiful / good heart fear
-----
Input:  cut temple chisel / adze stone brick deity god
Target: adze chisel cut stone brick temple god / deity
Pred:   adze chisel cut stone brick temple god / deity
-----
Input:  carries      pellet      man      handle      basket      grain
Target: man carries basket handle grain pellet
Pred:   man carries basket handle grain pellet
-----
Input:  temple peace altar above great god fire
Target: fire above peace altar god great temple
Pred:   fire above peace altar god great temple
-----
Input:  man plow hoe pellet grain field servant
Target: hoe plow field grain pellet man servant
Pred:   hoe plow field grain pellet man servant
-----
Input:  good writing book / scribe thing scroll scribe papyrus / beautiful
Target: scribe writing / scribe papyrus scroll book beautiful / good thing
Pred:   scribe writing / scribe book papyrus scroll beautiful / good thing
-----
Input:  head chick head swallow bird of owl belly animal quail of ox
Target: quail chick swallow owl head of bird head of ox animal belly
Pred:   quail chick swallow owl head of bird head of ox animal belly
-----
Input:  life (spirit) become (ba) to soul scarab emerge ka (ankh)
Target: to become scarab emerge life (ankh) soul (ba) ka (spirit)
Pred:   to become scarab emerge life (ankh) soul (ba) ka (spirit)
-----
Input:  see god horizon upon people
Target: people see god upon horizon
Pred:   people see god upon horizon
-----
Input:  he sedge / king mighty king authority / ring (king) power great
Target: king he / king sedge (king) authority power great / mighty ring
Pred:   king he / king sedge (king) authority power great / mighty ring
-----

Training complete. Best model saved in: ./best_model_checkpoint
```

<https://www.kaggle.com/code/sara1ahmed/bart-6-words-english>

7_WORDS_RESULTS



epoch	train_loss	val_loss	bleu	cosine
1	0.62835	0.095093	0.878128	0.985853
2	0.170217	0.036832	0.939898	0.995559
3	0.076021	0.022176	0.960297	0.997548
4	0.050775	0.018365	0.963384	0.99797
5	0.039959	0.01824	0.967917	0.997845
6	0.034575	0.015098	0.976452	0.998601
7	0.029588	0.012119	0.972396	0.998449
8	0.050766	0.013651	0.971876	0.998401
9	0.029516	0.015513	0.973213	0.998572
10	0.022456	0.017103	0.968438	0.997412
11	0.036196	0.011744	0.972342	0.997114
12	0.04303	0.013716	0.969913	0.99816
13	0.017011	0.007015	0.98182	0.998571
14	0.015838	0.01138	0.978346	0.998713
15	0.022109	0.014521	0.959261	0.995809
16	0.014357	0.010982	0.976933	0.997471
17	0.014262	0.012188	0.980019	0.998821

18	0.02304	0.01192	0.973929	0.996377
19	0.015807	0.007051	0.980676	0.99797
20	0.011343	0.008664	0.977651	0.998213
21	0.01594	0.011198	0.971872	0.994657
22	0.016688	0.010421	0.980212	0.998643
23	0.010637	0.008022	0.980852	0.99829
24	0.010948	0.00752	0.983731	0.999227
25	0.011389	0.005768	0.980022	0.997555
26	0.006984	0.00785	0.983235	0.99897
27	0.026673	0.010861	0.96488	0.994392
28	0.010891	0.010114	0.984265	0.999174
29	0.008252	0.011293	0.982717	0.998304
30	0.010707	0.012912	0.980134	0.99855
31	0.009063	0.008801	0.982702	0.998953
32	0.011357	0.009886	0.981005	0.997976
33	0.007125	0.008094	0.981864	0.998947
34	0.008255	0.011985	0.978748	0.997903
35	0.010074	0.014177	0.978329	0.998527
36	0.007087	0.007633	0.98466	0.99945
37	0.031403	0.007284	0.984871	0.999516
38	0.006146	0.007201	0.984681	0.999441
39	0.005567	0.007653	0.983199	0.997832
40	0.00602	0.008729	0.983048	0.998867
41	0.004525	0.009889	0.983895	0.999301
42	0.003845	0.01459	0.981381	0.998834
43	0.00528	0.01066	0.985639	0.999505
44	0.005667	0.009952	0.984444	0.998986

45	0.00617	0.011127	0.985097	0.999307
46	0.005872	0.019546	0.982745	0.999267
47	0.005108	0.011466	0.982776	0.998412
48	0.00549	0.014401	0.979323	0.997391
49	0.005781	0.010006	0.983661	0.999082
50	0.004979	0.012576	0.980716	0.998729
51	0.007231	0.008658	0.987023	0.999521
52	0.002297	0.010069	0.986871	0.999656
53	0.005451	0.013647	0.981298	0.998672
54	0.006366	0.014132	0.982911	0.998951
55	0.004958	0.012432	0.984532	0.999274
56	0.006208	0.014478	0.98287	0.999402
57	0.007295	0.016655	0.980205	0.998457
58	0.007943	0.012833	0.980955	0.997583
59	0.004179	0.016357	0.983495	0.9994
60	0.002096	0.016209	0.980046	0.998065
61	0.005774	0.015154	0.984514	0.999171
62	0.005421	0.014858	0.97838	0.997681
63	0.00423	0.012657	0.982394	0.998832
64	0.004788	0.010778	0.982453	0.998812
65	0.007731	0.01357	0.983235	0.998995
66	0.003802	0.013353	0.981955	0.998863
67	0.005053	0.011612	0.983548	0.999241
68	0.006356	0.012975	0.986853	0.999643
69	0.003842	0.01873	0.982864	0.998872
70	0.004502	0.010567	0.984744	0.999179
71	0.003818	0.015988	0.98199	0.998502

72	0.00268	0.014604	0.983957	0.999168
73	0.005486	0.017102	0.970942	0.994407
74	0.006413	0.012794	0.981908	0.998951
75	0.00277	0.015288	0.983007	0.998483
76	0.005845	0.014928	0.980581	0.998283
77	0.002148	0.014645	0.985092	0.999394
78	0.006221	0.020056	0.98167	0.998304
79	0.006444	0.01694	0.985011	0.999091
80	0.001945	0.014638	0.980529	0.997171
81	0.002701	0.018845	0.985685	0.999206
82	0.002862	0.013805	0.985649	0.999367
83	0.002273	0.019622	0.982647	0.998287
84	0.006388	0.013493	0.985668	0.999074
85	0.005231	0.015257	0.983402	0.998984
86	0.002712	0.015705	0.986324	0.999323
87	0.002061	0.012012	0.986833	0.999554
88	0.003853	0.014175	0.984442	0.999159
89	0.003948	0.01803	0.985712	0.998707
90	0.006939	0.014418	0.983742	0.99889
91	0.002514	0.014777	0.987703	0.99939
92	0.00162	0.014865	0.985158	0.999119
93	0.001267	0.015321	0.983465	0.999059
94	0.005037	0.017048	0.983128	0.999282
95	0.001006	0.016298	0.986463	0.999325
96	0.004283	0.018202	0.983854	0.998824
97	0.003257	0.020973	0.979577	0.9985
98	0.004953	0.01962	0.980864	0.997872

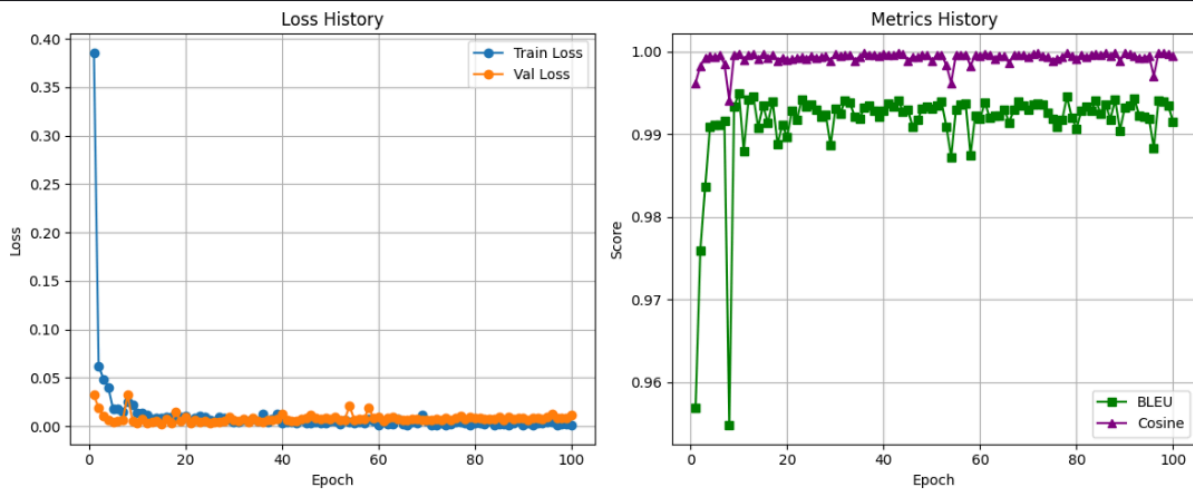
99	0.003802	0.017131	0.979953	0.998847
100	0.004397	0.017364	0.984264	0.99931

```
=== SAMPLE RECONSTRUCTIONS (BEST MODEL) ===
Input:  god wisdom hears man old of
Target: old man hears wisdom of god
Pred:   old man hears wisdom of god
-----
Input:  lord upon above throne the the sits
Target: the lord sits upon the throne above
Pred:   the lord sits upon the throne above
-----
Input:  truth feather sees heart (truth/maat) eye
Target: eye sees truth feather (truth/maat) heart
Pred:   eye sees truth feather (truth/maat) heart
-----
Input:  the in the jars sees lord rack
Target: the lord sees the jars in rack
Pred:   the lord sees the jars in rack
-----
Input:  content love life satisfied people peace
Target: people love peace content satisfied life
Pred:   people love peace content satisfied life
-----
Input:  gives youth the milk jar father a
Target: the father gives a milk jar youth
Pred:   the father gives a milk jar youth
-----
Input:  life lake river ripple water field
Target: river lake water ripple life field
Pred:   lake river water ripple life field
-----
Input:  the the sees king field grain sedge
Target: the sedge king sees the field grain
Pred:   the sedge king sees the field grain
-----
Input:  emerge power life underground (ba) west god soul
Target: underground west soul (ba) life god power emerge
Pred:   underground west soul (ba) life god power emerge
-----
Input:  plant sickle a uses a man on
Target: a man uses a sickle on plant
Pred:   a man uses a sickle on plant
-----

Training complete. Best model saved in: ./best_model_checkpoint
```

<https://www.kaggle.com/code/soliharhossam/bart-7-words-english>

8_WORDS_RESULTS



epoch	train_loss	val_loss	bleu	cosine
1	0.385809	0.032103	0.956863	0.9962
2	0.06229	0.018824	0.97589	0.998245
3	0.048631	0.010398	0.983571	0.999151
4	0.040089	0.006386	0.99091	0.999415
5	0.017748	0.004753	0.991136	0.999284
6	0.017566	0.005371	0.991113	0.999539
7	0.014472	0.006603	0.991589	0.998506
8	0.025135	0.032549	0.954758	0.994103
9	0.021644	0.005838	0.993285	0.999633
10	0.013661	0.002701	0.994901	0.99969
11	0.01351	0.007008	0.987948	0.998918
12	0.011113	0.003538	0.994154	0.999598
13	0.007851	0.003774	0.994518	0.999646
14	0.008406	0.004917	0.990772	0.999037
15	0.008889	0.002545	0.993459	0.999649
16	0.009138	0.00739	0.991316	0.999216
17	0.010015	0.0035	0.993907	0.999569

18	0.008393	0.014709	0.988745	0.998861
19	0.009929	0.005208	0.991089	0.999265
20	0.010334	0.009437	0.989668	0.99892
21	0.005263	0.003033	0.992848	0.999083
22	0.008382	0.006804	0.991691	0.999163
23	0.01006	0.00479	0.99415	0.999319
24	0.009676	0.005242	0.99326	0.999144
25	0.006359	0.003394	0.993564	0.99946
26	0.005064	0.004177	0.9929	0.999264
27	0.009113	0.004633	0.992148	0.999273
28	0.008038	0.005134	0.992394	0.999471
29	0.006126	0.009833	0.988702	0.99888
30	0.004694	0.006522	0.993111	0.999693
31	0.00413	0.005677	0.992469	0.999471
32	0.006141	0.006937	0.994066	0.999597
33	0.005626	0.004702	0.993755	0.999525
34	0.00712	0.008344	0.992144	0.998891
35	0.005153	0.005727	0.991852	0.999373
36	0.01224	0.004602	0.993151	0.999776
37	0.004175	0.006101	0.993421	0.99963
38	0.007199	0.006128	0.992848	0.999608
39	0.012688	0.007837	0.992124	0.999496
40	0.003521	0.012941	0.992856	0.999696
41	0.00415	0.006748	0.993656	0.999619
42	0.004708	0.005753	0.99336	0.999542
43	0.003526	0.005755	0.994093	0.999835
44	0.006214	0.00765	0.99275	0.999676

45	0.003326	0.008171	0.992975	0.998893
46	0.003294	0.011221	0.990898	0.999334
47	0.004146	0.008187	0.991696	0.999292
48	0.003008	0.006992	0.993029	0.999561
49	0.002898	0.008719	0.993287	0.99952
50	0.005206	0.007177	0.993021	0.998792
51	0.005102	0.009118	0.993474	0.999565
52	0.001701	0.006108	0.993895	0.999629
53	0.005982	0.00674	0.990912	0.998328
54	0.004679	0.020798	0.987174	0.996172
55	0.003732	0.006825	0.992993	0.999594
56	0.004786	0.007623	0.993595	0.999632
57	0.002884	0.007502	0.993668	0.999637
58	0.007512	0.019198	0.987465	0.998256
59	0.004923	0.007987	0.992232	0.999554
60	0.001484	0.009775	0.991878	0.999502
61	0.003745	0.005833	0.993853	0.999701
62	0.001984	0.008117	0.991934	0.999621
63	0.002407	0.009522	0.992263	0.999081
64	0.007706	0.007554	0.992269	0.999448
65	0.002496	0.005904	0.993012	0.99947
66	0.000925	0.006305	0.991408	0.998563
67	0.004167	0.007187	0.992921	0.999598
68	0.003283	0.00698	0.993879	0.999546
69	0.011103	0.006785	0.993445	0.99959
70	0.005409	0.006005	0.992935	0.999359
71	0.000768	0.006281	0.993518	0.999723

72	0.000859	0.007767	0.993652	0.999764
73	0.002836	0.006643	0.993603	0.999437
74	0.001032	0.008522	0.992588	0.999351
75	0.002457	0.006364	0.991873	0.998884
76	0.003823	0.008875	0.990812	0.999124
77	0.004446	0.010656	0.99168	0.999283
78	0.003275	0.007426	0.994543	0.999857
79	0.001549	0.008985	0.991985	0.999459
80	0.003057	0.008943	0.990627	0.999072
81	0.003131	0.008697	0.992873	0.999585
82	0.001909	0.007702	0.993281	0.99937
83	0.005032	0.0074	0.992859	0.999545
84	0.001565	0.007501	0.994077	0.999698
85	0.002159	0.009045	0.992451	0.999528
86	0.002485	0.006772	0.993618	0.999827
87	0.001063	0.00904	0.991772	0.999465
88	0.002926	0.007033	0.994119	0.999817
89	0.0053	0.008622	0.990408	0.998807
90	0.000833	0.008572	0.993255	0.999796
91	0.003136	0.006793	0.993421	0.999653
92	0.001588	0.008243	0.994283	0.999513
93	0.002901	0.007996	0.99218	0.999164
94	0.003359	0.007674	0.992117	0.999261
95	0.004415	0.009138	0.991898	0.999432
96	0.003759	0.012473	0.988298	0.997018
97	0.001076	0.008465	0.994093	0.999829
98	0.001725	0.008741	0.993918	0.999812

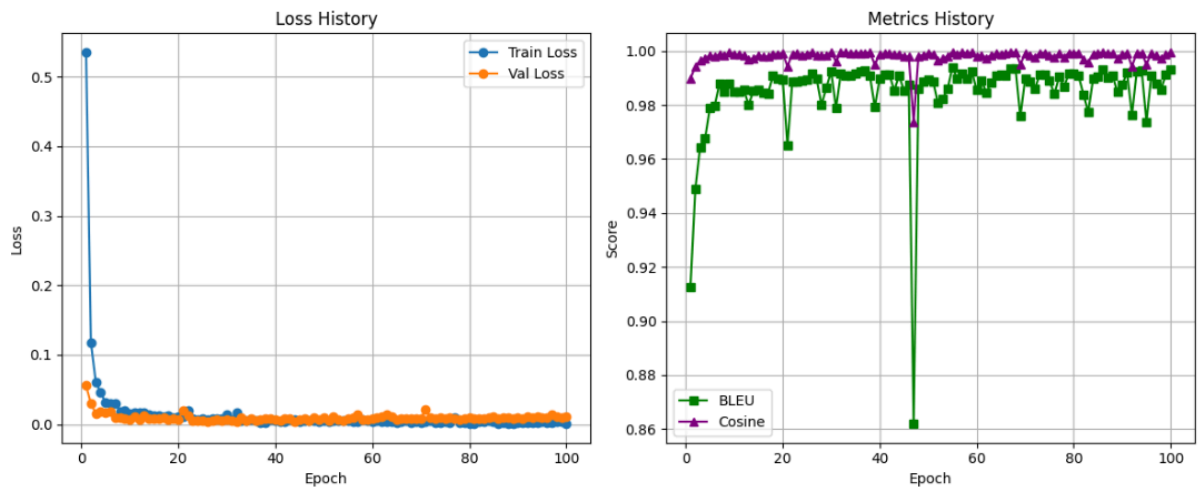
99	0.002378	0.00827	0.99343	0.999711
100	0.001117	0.011659	0.991532	0.999459

```
=== SAMPLE RECONSTRUCTIONS (BEST MODEL) ===
Input:  the master the sees all of truth people
Target: the master of all people sees the truth
Pred:   the master of all people sees the truth
-----
Input:  the thousand river stars above bank a shine
Target: a thousand stars shine above the river bank
Pred:   a thousand stars shine above the river bank
-----
Input:  the hides plant desert in hare green the
Target: the desert hare hides in the green plant
Pred:   the desert hare hides in the green plant
-----
Input:  writes book truth about the scribe a the
Target: the scribe writes a book about the truth
Pred:   the scribe writes a book about the truth
-----
Input:  man old finds in peace city the the
Target: the old man finds peace in the city
Pred:   the old man finds peace in the city
-----
Input:  studies fire papyrus scroll light calmly scribe by
Target: scribe studies papyrus scroll by fire light calmly
Pred:   scribe studies papyrus scroll scroll by fire light calmly
-----
Input:  with handle woman basket house carries to
Target: woman carries basket with handle to house
Pred:   woman carries basket with handle to house
-----
Input:  the the green hare desert in plant hides
Target: the desert hare hides in the green plant
Pred:   the desert hare hides in the green plant
-----
Input:  gate woman to basket with a walks a
Target: a woman with a basket walks to gate
Pred:   a woman with a basket walks to gate
-----
Input:  woman calmly road toward follows riverbank west
Target: woman follows riverbank road toward west calmly
Pred:   woman follows riverbank road toward west calmly
-----

Training complete. Best model saved in: ./best_model_checkpoint
```

<https://www.kaggle.com/code/asd4zxc/bart-8-word-s-english>

9_WORDS_RESULTS



epoch	train_loss	val_loss	bleu	cosine
1	0.535598	0.055684	0.912547	0.989777
2	0.11806	0.030091	0.948887	0.99442
3	0.061057	0.015736	0.964461	0.996418
4	0.046103	0.018375	0.967506	0.997311
5	0.031791	0.016281	0.978789	0.998374
6	0.02957	0.018505	0.979849	0.997993
7	0.030418	0.0092	0.988018	0.998892
8	0.018663	0.009169	0.984904	0.998404
9	0.019294	0.007792	0.987817	0.999376
10	0.016075	0.007282	0.985049	0.99885
11	0.017093	0.010573	0.985018	0.998762
12	0.016501	0.006263	0.985505	0.998411
13	0.016321	0.012141	0.980091	0.996835
14	0.013627	0.008874	0.98539	0.997377
15	0.0129	0.007678	0.98565	0.998485
16	0.012262	0.007953	0.984476	0.998042
17	0.010576	0.009572	0.984298	0.997979

18	0.013304	0.007293	0.990863	0.998901
19	0.009965	0.007914	0.989923	0.998922
20	0.011521	0.007212	0.989537	0.999203
21	0.011613	0.019628	0.964958	0.994297
22	0.019319	0.013122	0.988852	0.99899
23	0.00845	0.005698	0.988583	0.999065
24	0.007468	0.005975	0.989086	0.998612
25	0.00811	0.005529	0.9896	0.99892
26	0.007246	0.003904	0.991622	0.99947
27	0.008113	0.005063	0.989886	0.998525
28	0.007706	0.00626	0.980102	0.998257
29	0.008773	0.004993	0.98638	0.998492
30	0.013382	0.006261	0.992392	0.999489
31	0.006606	0.004884	0.978795	0.99613
32	0.017392	0.004458	0.991658	0.99954
33	0.004935	0.009655	0.990955	0.999465
34	0.004837	0.004848	0.990995	0.99901
35	0.006688	0.008579	0.991732	0.999108
36	0.005086	0.004791	0.992345	0.999336
37	0.002862	0.006217	0.99288	0.999353
38	0.003036	0.00788	0.990766	0.999449
39	0.006904	0.008485	0.979228	0.994973
40	0.005709	0.006256	0.989676	0.998756
41	0.00406	0.005362	0.991257	0.99902
42	0.003285	0.00756	0.991352	0.999145
43	0.007071	0.008639	0.985376	0.998701
44	0.007313	0.004028	0.990889	0.998987

45	0.00432	0.007402	0.985406	0.998091
46	0.00571	0.007716	0.987574	0.998018
47	0.005402	0.00596	0.861818	0.973845
48	0.005322	0.009795	0.986032	0.998145
49	0.004064	0.006646	0.988565	0.998526
50	0.00495	0.01018	0.989418	0.999141
51	0.00437	0.004992	0.988798	0.998938
52	0.005541	0.011746	0.98091	0.996451
53	0.005634	0.007163	0.982492	0.997136
54	0.005036	0.005498	0.985895	0.998227
55	0.005293	0.008126	0.993934	0.999662
56	0.00402	0.010281	0.989719	0.998767
57	0.003598	0.014704	0.991592	0.99959
58	0.007387	0.007387	0.989839	0.99915
59	0.004726	0.00667	0.992391	0.999429
60	0.004438	0.008497	0.985583	0.998133
61	0.005011	0.010299	0.989179	0.998525
62	0.003922	0.011323	0.984442	0.997457
63	0.00336	0.013513	0.988402	0.998542
64	0.003218	0.010814	0.990776	0.999049
65	0.00276	0.007328	0.991264	0.998941
66	0.003314	0.007753	0.990918	0.999078
67	0.005195	0.007864	0.993378	0.999579
68	0.003103	0.007818	0.99356	0.999605
69	0.003397	0.007617	0.975926	0.995177
70	0.003115	0.008921	0.989891	0.999079
71	0.004031	0.021122	0.988689	0.998376

72	0.004747	0.009219	0.986064	0.99779
73	0.002557	0.00788	0.991271	0.999365
74	0.002386	0.009222	0.991333	0.99927
75	0.004412	0.008727	0.989049	0.998591
76	0.002021	0.010313	0.984043	0.997744
77	0.009042	0.008474	0.99041	0.99921
78	0.002293	0.007106	0.986841	0.99777
79	0.002352	0.008131	0.991521	0.999324
80	0.000481	0.010118	0.991862	0.999357
81	0.001134	0.00841	0.990841	0.999322
82	0.003917	0.008422	0.98385	0.997028
83	0.003735	0.008852	0.977511	0.99591
84	0.004764	0.009582	0.989884	0.998982
85	0.004182	0.010999	0.990589	0.999366
86	0.001355	0.008006	0.993132	0.999446
87	0.00237	0.00935	0.990211	0.99924
88	0.001377	0.009516	0.990893	0.999048
89	0.001623	0.00763	0.98484	0.997437
90	0.002111	0.009098	0.987493	0.998803
91	0.003132	0.008494	0.99207	0.999272
92	0.002691	0.011545	0.97644	0.994346
93	0.002258	0.009408	0.992464	0.999302
94	0.001962	0.011498	0.992942	0.999329
95	0.003294	0.009876	0.97385	0.995052
96	0.002235	0.009697	0.991086	0.999273
97	0.001976	0.014011	0.988034	0.998285
98	0.003408	0.011045	0.985818	0.997344

99	0.004605	0.009551	0.991341	0.99867
100	0.001241	0.010482	0.993352	0.99946

```

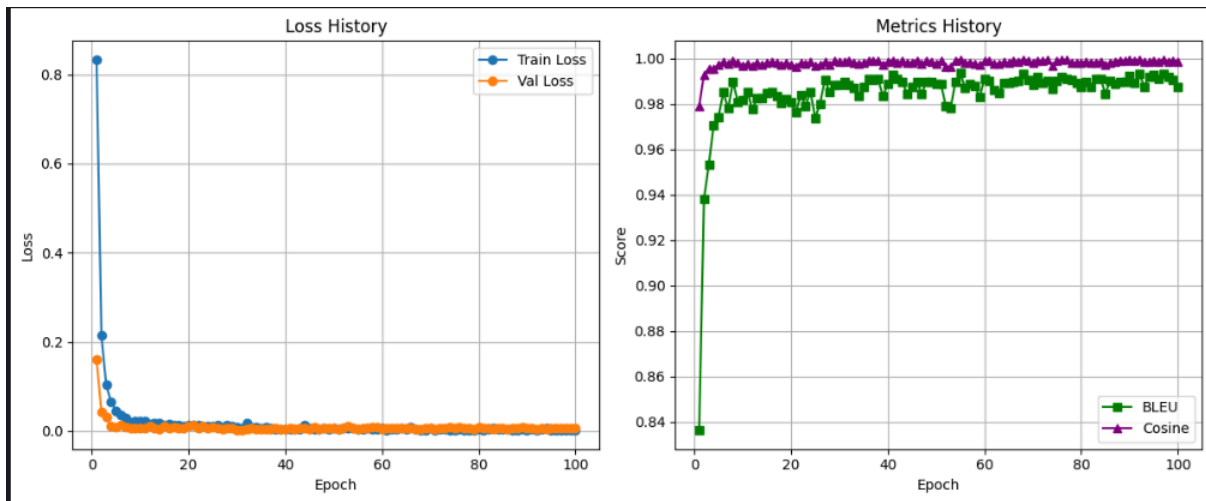
=== SAMPLE RECONSTRUCTIONS (BEST MODEL) ===
Input:  like field / / man plows house (tills) stone (or back father example) behind as
Target: man plows (tills) field like / as (or example) father behind / back stone house
Pred:   man plows (tills) field like / as (or example) father behind / back stone house
-----
Input:  man city old road youth become behind town back / temple stone
Target: youth become old man behind / back stone temple city town road
Pred:   youth become old man behind / back stone temple city town road
-----
Input:  behind sees chisel / adze back above king stone temple
Target: king sees adze chisel above stone behind / back temple
Pred:   king sees adze chisel above stone behind / back temple
-----
Input:  peace give / life beautiful king behind to back (ankh) / good altar /
Target: king to give life (ankh) behind / back beautiful / good peace / altar
Pred:   king to give life (ankh) behind / back beautiful / good peace / altar
-----
Input:  heart beautiful life / good hear god above listen / / deity
Target: god / deity hear / listen heart above beautiful / good life
Pred:   god / deity hear / listen heart above beautiful / good life
-----
Input:  servant king grant offering reward
Target: king grant servant reward offering
Pred:   king grant servant reward offering
-----
Input:  town road adze scribe brick chisel stone sees house above
Target: scribe sees adze chisel above brick stone house town road
Pred:   scribe sees adze chisel above brick stone house town road
-----
Input:  be) hill sees behind / back emerge road hare man (to desert desert
Target: man sees desert hare (to be) emerge behind / back desert hill road
Pred:   man sees desert hare (to be) emerge behind / back desert hill road
-----
Input:  book house (lord / basket back above scribe town behind sees all) / stone
Target: scribe sees book above basket (lord / all) behind / back stone house town
Pred:   scribe sees book above basket (lord / all) behind / back stone house town
-----
Input:  mat mother stool sit child
Target: mother sit stool mat child
Pred:   mother sit stool mat child
-----

Training complete. Best model saved in: ./best_model_checkpoint

```

<https://www.kaggle.com/code/ahmedkhaledeissa/art-9-words-english>

10_WORDS_RESULTS



epoch	train_loss	val_loss	bleu	cosine
1	0.833499	0.161712	0.836098	0.979035
2	0.213977	0.042113	0.938094	0.992943
3	0.103077	0.031735	0.953359	0.995709
4	0.065172	0.010576	0.970715	0.99555
5	0.045796	0.00837	0.974309	0.997395
6	0.036702	0.012463	0.985075	0.998503
7	0.029269	0.008111	0.977986	0.997604
8	0.021373	0.006768	0.989821	0.998946
9	0.02234	0.005958	0.980832	0.998356
10	0.022185	0.005802	0.981883	0.99694
11	0.022225	0.006338	0.985419	0.997827
12	0.015942	0.011871	0.977582	0.996977
13	0.018152	0.007315	0.982635	0.998265
14	0.017136	0.00417	0.982488	0.997125
15	0.014271	0.010508	0.984663	0.99818
16	0.014892	0.006923	0.985298	0.998429
17	0.014455	0.008375	0.983497	0.998207

18	0.012936	0.007729	0.980281	0.997313
19	0.012218	0.005978	0.982137	0.998197
20	0.013647	0.012318	0.980719	0.99748
21	0.012831	0.012888	0.976229	0.996554
22	0.014328	0.007456	0.984039	0.998145
23	0.010749	0.010885	0.979256	0.997825
24	0.011354	0.007755	0.98549	0.998416
25	0.010771	0.008016	0.973794	0.996935
26	0.013012	0.007108	0.980151	0.997358
27	0.008253	0.00485	0.990699	0.998431
28	0.01337	0.007463	0.985165	0.997321
29	0.012193	0.007025	0.988504	0.998995
30	0.008555	0.002835	0.988233	0.998481
31	0.005781	0.002687	0.98978	0.998518
32	0.017121	0.004642	0.988229	0.999182
33	0.008218	0.006619	0.98691	0.998284
34	0.007946	0.003971	0.983687	0.997744
35	0.007869	0.00537	0.987377	0.998206
36	0.009933	0.003522	0.991191	0.999026
37	0.007216	0.004276	0.990595	0.99891
38	0.004813	0.007493	0.990993	0.999079
39	0.005006	0.004826	0.983431	0.997129
40	0.004532	0.005413	0.988849	0.998687
41	0.004372	0.0065	0.992851	0.999272
42	0.004997	0.005163	0.991081	0.998392
43	0.003725	0.006969	0.989731	0.998941
44	0.013303	0.006265	0.98445	0.998097

45	0.006925	0.00468	0.987442	0.998413
46	0.004256	0.008462	0.989787	0.999297
47	0.004406	0.004986	0.984547	0.997556
48	0.007255	0.007621	0.989601	0.998993
49	0.004827	0.006883	0.989765	0.998718
50	0.006586	0.006196	0.988905	0.997813
51	0.003774	0.004941	0.988944	0.99892
52	0.00585	0.007242	0.979045	0.996237
53	0.007795	0.010209	0.978384	0.996621
54	0.005966	0.006617	0.989783	0.99923
55	0.003477	0.004828	0.993698	0.999654
56	0.0039	0.006028	0.986864	0.998194
57	0.007515	0.007747	0.988876	0.998338
58	0.00502	0.008254	0.988041	0.997764
59	0.003476	0.010042	0.983137	0.997156
60	0.006505	0.004817	0.99122	0.998951
61	0.002753	0.006074	0.990067	0.99903
62	0.004332	0.00621	0.986339	0.997687
63	0.004094	0.006594	0.984978	0.997746
64	0.006125	0.006082	0.989348	0.998631
65	0.003774	0.006613	0.989338	0.998122
66	0.007942	0.007159	0.989742	0.999093
67	0.003389	0.004817	0.990008	0.998844
68	0.002527	0.004444	0.993311	0.999473
69	0.002857	0.007848	0.990432	0.999197
70	0.006008	0.00612	0.98847	0.998152
71	0.003041	0.005001	0.991754	0.999133

72	0.005007	0.007121	0.989322	0.998893
73	0.005623	0.00629	0.990138	0.999325
74	0.003219	0.008013	0.986589	0.997011
75	0.001949	0.006106	0.990188	0.998928
76	0.004736	0.008802	0.991742	0.999429
77	0.003133	0.007734	0.991181	0.99955
78	0.001745	0.007627	0.9906	0.99821
79	0.002557	0.005353	0.98926	0.998221
80	0.003431	0.008865	0.98765	0.998331
81	0.003315	0.006198	0.989627	0.998599
82	0.004328	0.007494	0.98756	0.998263
83	0.006464	0.005588	0.991269	0.998328
84	0.003514	0.006486	0.99125	0.998815
85	0.005005	0.005717	0.98447	0.997187
86	0.003898	0.004836	0.990347	0.998167
87	0.001982	0.005945	0.988799	0.998807
88	0.001459	0.005778	0.989774	0.999224
89	0.004167	0.008368	0.989821	0.999086
90	0.001501	0.006373	0.99255	0.999373
91	0.003903	0.00683	0.989416	0.999197
92	0.00262	0.005143	0.993512	0.999457
93	0.004427	0.006191	0.987702	0.998535
94	0.006282	0.00494	0.992118	0.998698
95	0.002619	0.006746	0.992844	0.998934
96	0.002472	0.006975	0.990981	0.99873
97	0.00249	0.006161	0.99309	0.999652
98	0.00281	0.006298	0.991858	0.998667

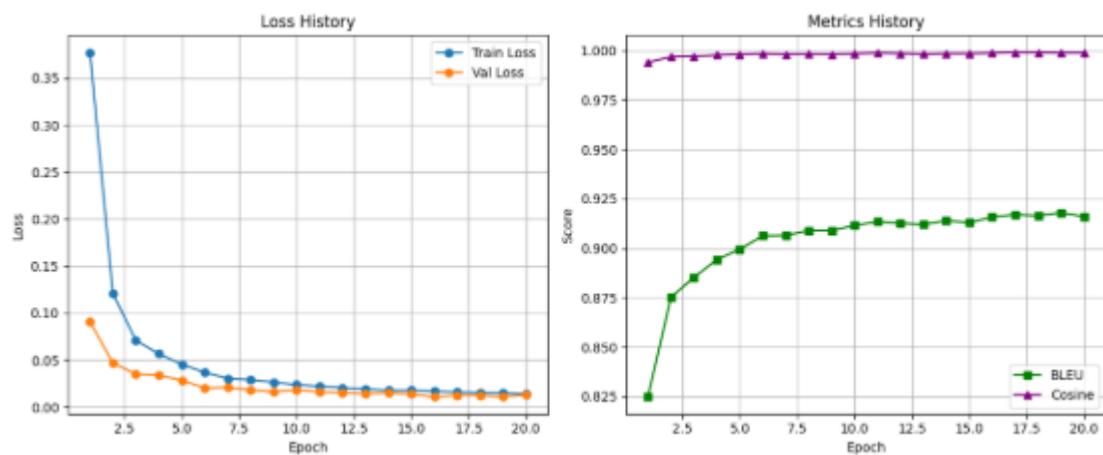
99	0.002306	0.006486	0.99049	0.999231
100	0.001673	0.006473	0.987588	0.998853

```
=== SAMPLE RECONSTRUCTIONS (BEST MODEL) ===
Input: scroll the leaf scribe of hand reed papyrus hold for
Target: hand of the scribe hold reed leaf for papyrus scroll
Pred: hand of the scribe hold reed leaf for papyrus scroll
-----
Input: sky star wonder night life thousand beauty
Target: night sky star thousand beauty wonder life
Pred: night sky star thousand beauty wonder life
-----
Input: hold vessel the great stone the wall tethering rope near
Target: tethering rope hold the vessel near the great stone wall
Pred: tethering rope hold the vessel near the great stone wall
-----
Input: and clump chick fly lake quail papyrus above the swallow
Target: quail chick and swallow fly above the papyrus clump lake
Pred: quail chick and swallow fly above the papyrus clump lake
-----
Input: hear seated heart behind man master gate all of of
Target: master of all hear heart of seated man behind gate
Pred: master of all hear heart of seated man behind gate
-----
Input: milk stay in loaf jar shelter reed and the conical
Target: conical loaf and milk jar stay in the reed shelter
Pred: conical loaf and milk jar stay in the reed shelter
-----
Input: god desert stay the above hill of great mighty authority
Target: mighty authority of god stay above the great desert hill
Pred: mighty authority of god stay above the great desert hill
-----
Input: do god hear and truth of of scribe mouth writing
Target: scribe hear mouth of god and do writing of truth
Pred: scribe hear mouth of god and do writing of truth
-----
Input: the bird soul ba mountain fly father above like of
Target: soul ba of father fly above the mountain like bird
Pred: soul ba of father fly above the mountain like bird
-----
Input: stay rib altar offering belly on the animal side and
Target: animal belly and side rib stay on the offering altar
Pred: animal belly and side rib stay on the offering altar
-----

Training complete. Best model saved in: ./best_model_checkpoint
```

<https://www.kaggle.com/code/ahmedkhaled43/bart-10-words-english>

All_words_english



epoch	train_loss	val_loss	bleu	cosine
1	0.377043	0.090675	0.824745	0.993963
2	0.120747	0.046974	0.875055	0.996817
3	0.070977	0.035158	0.885182	0.997048
4	0.05664	0.03389	0.894121	0.997684
5	0.045589	0.028588	0.899493	0.997965
6	0.036897	0.02034	0.906221	0.998432
7	0.030606	0.021007	0.906405	0.99804
8	0.029082	0.01829	0.908855	0.99826
9	0.026474	0.016364	0.908816	0.998099
10	0.023748	0.018279	0.911467	0.998439
11	0.022186	0.016244	0.913383	0.998722
12	0.020595	0.015561	0.9126	0.998548
13	0.01933	0.014065	0.911949	0.998226
14	0.017865	0.01502	0.913909	0.998473
15	0.017904	0.013818	0.91289	0.998476
16	0.016821	0.010948	0.915786	0.998685
17	0.016068	0.012458	0.916934	0.998978
18	0.015404	0.012263	0.916383	0.998899

19	0.01506	0.011171	0.917956	0.998756
20	0.014012	0.012885	0.915947	0.998776

```
=== SAMPLE RECONSTRUCTIONS (BEST MODEL) ===
Input:  servant plows road
Target: servant plows road
Pred:   servant plows road
-----
Input:  holds power city throne authority town / seated people temple king
Target: seated king holds authority power throne temple people city / town
Pred:   seated king holds authority power throne temple people city / town
-----
Input:  mountain desert near
Target: mountain near desert
Pred:   mountain near desert
-----
Input:  live people riverbank lake field
Target: people live riverbank lake field
Pred:   people live riverbank lake field
-----
Input:  viper hide basket the with in horned handle near brick
Target: horned viper hide in the basket with handle near brick
Pred:   horned viper hide in the basket with handle near brick
-----
Input:  scroll above stool book / scribe papyrus scribe temple writing mat /
Target: scribe writing / scribe papyrus scroll book above stool / mat temple
Pred:   scribe writing / scribe papyrus scroll book above stool / mat temple
-----
Input:  stone house good beautiful sees / above horizon king great
Target: king sees great stone house above beautiful / good horizon
Pred:   king sees great stone house above beautiful / good horizon
-----
Input:  life mother give soul
Target: mother give soul life
Pred:   mother give soul life
-----
Input:  offerings temple scroll on inside papyrus scribe records
Target: scribe records offerings on papyrus scroll inside temple
Pred:   scribe records offerings on papyrus scroll inside temple
-----
Input:  (ankh) youth life marrow old venerated man / father satisfy peace /
Target: old man venerated / marrow father youth life (ankh) peace / satisfy
Pred:   old man venerated / marrow father youth life (ankh) peace / satisfy
-----

Training complete. Best model saved in: ./best_model_checkpoint
```

<https://www.kaggle.com/code/khaledeissa1/all-words-words-english>



1. Training Dynamics (Learning Behavior)

These show how the model learned over time.

A. Loss Curves

From:

```
history['train_loss']  
history['val_loss']
```

Report:

- Training Loss vs Epoch
- Validation Loss vs Epoch
- Epoch where **best validation loss** occurred
- Value of **best validation loss**

Why it matters: Shows convergence, overfitting/underfitting.

B. Metric Curves

From:

```
history['bleu']  
history['cosine'] # if enabled
```

Report:

- BLEU score vs Epoch
- Cosine similarity vs Epoch (if calculated)
- **Best BLEU score** achieved
- Epoch of best BLEU

Why it matters: Shows actual reconstruction quality, not just loss.

2. Best Model Information

From checkpoint logic:

```
if avg_val_loss < best_val_loss:  
    model.save_pretrained(SAVE_DIR)
```

Report:

- Model used: **facebook/bart-base**
 - Best model selected based on: **Lowest Validation Loss**
 - Path of saved best model:
`./best_model_checkpoint`
 - Epoch number when best model was saved
-

3. Final Evaluation Metrics (Validation Set)

From:

```
bleu, cosine =  
calculate_metrics(predictions,  
references)
```

Report final values (from best epoch or last epoch depending on your choice):

- Final Validation BLEU Score (mean)

- Final Validation Cosine Similarity (mean, if used)
- Final Validation Loss

Put these in a clean table:

Metric	Value
Validation Loss	X.XX XX
BLEU Score	X.XX XX
Cosine Similarity	X.XX XX

4. Qualitative Results (Sample Reconstructions)

From:

```
print("=== SAMPLE RECONSTRUCTIONS  
===")
Input / Target / Pred
```

Report:

Include **5–10 example reconstructions** in a table:

Shuffled Input	Ground Truth Sentence	Model Prediction
----------------	-----------------------	------------------

Why it matters: Shows real behavior, fluency, and common errors.

5. Saved Experimental Artifacts

Mention these files in your report:

File	Contents
<code>training_metrics.csv</code>	Epoch-wise loss and metric history
<code>reconstruction_results.csv</code>	Full validation predictions
<code>./best_model_checkpoint/</code>	Final trained model + tokenizer

⚙️ 6. Experimental Setup Summary (from config)

Always required in reports:

Parameter	Value
Model	BART-base
Max sequence length	32
Batch size	16
Epochs	100
Learning rate	5e-5
Optimizer	AdamW
Scheduler	None
Augmentation	Word shuffling (10 variants per sentence)
Train/Val split	80% / 20%

🧠 7. Dataset Statistics

From earlier cells but essential for reporting:

- Total dataset size after augmentation
 - Number of training samples
 - Number of validation samples
 - Average sentence length (optional but good)
-

Minimum Must-Have Figures for the Report

- 1. Loss vs Epoch graph**
- 2. BLEU (and Cosine) vs Epoch graph**
- 3. Table of final metrics**
- 4. Table of sample reconstructions**